



```
In [23... import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [24... df = pd.read_csv(r'C:\Users\SANDRA\Downloads\archive\movie_profit.csv')
```

```
In [27... df.head()
```

```
Out[27]:
```

|   | Unnamed: 0 | release_date | movie                            | production_budget | domestic_gross | worldwide_gross |
|---|------------|--------------|----------------------------------|-------------------|----------------|-----------------|
| 0 | 1          | 6/22/2007    | Evan Almighty                    | 175000000.0       | 100289690.0    | 1.741313e+08    |
| 1 | 2          | 7/28/1995    | Waterworld                       | 175000000.0       | 88246220.0     | 2.642462e+08    |
| 2 | 3          | 5/12/2017    | King Arthur: Legend of the Sword | 175000000.0       | 39175066.0     | 1.399507e+08    |
| 3 | 4          | 12/25/2013   | 47 Ronin                         | 175000000.0       | 38362475.0     | 1.517168e+08    |
| 4 | 5          | 6/22/2018    | Jurassic World: Fallen Kingdom   | 170000000.0       | 416769345.0    | 1.304866e+09    |

```
In [7]: df.tail()
```

```
Out[7]:
```

|      | Unnamed: 0 | release_date | movie                       | production_budget | domestic_gross | worldwide_g |
|------|------------|--------------|-----------------------------|-------------------|----------------|-------------|
| 3396 | 3397       | 10/28/2011   | Like Crazy                  | 250000.0          | 3395391.0      | 37284       |
| 3397 | 3398       | 7/14/2000    | Chuck&Buck                  | 250000.0          | 1055671.0      | 11576       |
| 3398 | 3399       | 3/28/1997    | Love and Other Catastrophes | 250000.0          | 212285.0       | 7432        |
| 3399 | 3400       | 8/28/1998    | I Married a Strange Person  | 250000.0          | 203134.0       | 2031        |
| 3400 | 3401       | 7/22/2005    | November                    | 250000.0          | 191862.0       | 1918        |

```
In [28... df.shape
```

```
Out[28]: (3401, 9)
```

```
In [29... df.columns
```

```
Out[29]: Index(['Unnamed: 0', 'release_date', 'movie', 'production_budget',
              'domestic_gross', 'worldwide_gross', 'distributor', 'mpaa_rating',
              'genre'],
              dtype='str')
```

```
In [30]: df.drop(columns = ('Unnamed: 0'), inplace = True)
```

```
In [31]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3401 entries, 0 to 3400
Data columns (total 8 columns):
#   Column                      Non-Null Count  Dtype
---  -
0   release_date                3401 non-null   str
1   movie                       3401 non-null   str
2   production_budget           3401 non-null   float64
3   domestic_gross              3401 non-null   float64
4   worldwide_gross             3401 non-null   float64
5   distributor                  3353 non-null   str
6   mpaa_rating                  3264 non-null   str
7   genre                       3401 non-null   str
dtypes: float64(3), str(5)
memory usage: 212.7 KB
```

```
In [33]: df.isnull().sum()
```

```
Out[33]: release_date      0
movie                    0
production_budget        0
domestic_gross           0
worldwide_gross          0
distributor              48
mpaa_rating              137
genre                    0
dtype: int64
```

```
In [34]: df.dropna(inplace = True)
```

```
In [36]: df.isna().sum()
```

```
Out[36]: release_date      0
movie                    0
production_budget        0
domestic_gross           0
worldwide_gross          0
distributor              0
mpaa_rating              0
genre                    0
dtype: int64
```

```
In [39]: df.duplicated().sum()
```

```
Out[39]: np.int64(0)
```

```
df.describe()
```

```
In [40]: df.sort_values('production_budget', ascending = True)
```

```
Out[40]:
```

|             | release_date | movie                            | production_budget | domestic_gross | worldwide_gross |      |
|-------------|--------------|----------------------------------|-------------------|----------------|-----------------|------|
| <b>3400</b> | 7/22/2005    | November                         | 250000.0          | 191862.0       | 191862.0        | Sony |
| <b>3398</b> | 3/28/1997    | Love and Other Catastrophes      | 250000.0          | 212285.0       | 743216.0        |      |
| <b>3397</b> | 7/14/2000    | Chuck&Buck                       | 250000.0          | 1055671.0      | 1157672.0       |      |
| <b>3396</b> | 10/28/2011   | Like Crazy                       | 250000.0          | 3395391.0      | 3728400.0       | Par  |
| <b>3395</b> | 4/11/2003    | Better Luck Tomorrow             | 250000.0          | 3802390.0      | 3809226.0       | Par  |
| ...         | ...          | ...                              | ...               | ...            | ...             |      |
| <b>14</b>   | 5/27/2016    | Alice Through the Looking Glass  | 170000000.0       | 77042381.0     | 276934087.0     |      |
| <b>2</b>    | 5/12/2017    | King Arthur: Legend of the Sword | 175000000.0       | 39175066.0     | 139950708.0     |      |
| <b>3</b>    | 12/25/2013   | 47 Ronin                         | 175000000.0       | 38362475.0     | 151716815.0     |      |
| <b>0</b>    | 6/22/2007    | Evan Almighty                    | 175000000.0       | 100289690.0    | 174131329.0     |      |
| <b>1</b>    | 7/28/1995    | Waterworld                       | 175000000.0       | 88246220.0     | 264246220.0     |      |

3230 rows × 8 columns

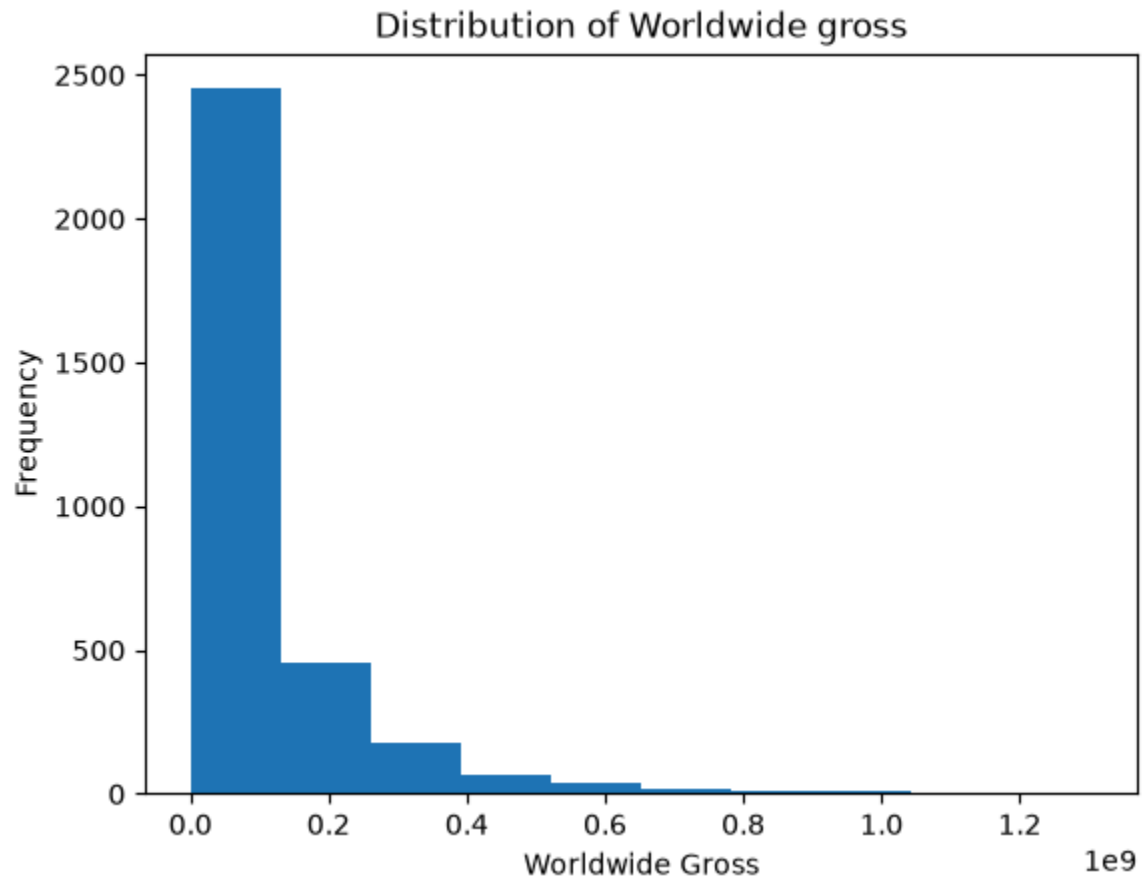
```
In [41]: df['genre'].value_counts()
```

```
Out[41]: genre
Drama      1180
Comedy      775
Action      540
Adventure   469
Horror      266
Name: count, dtype: int64
```

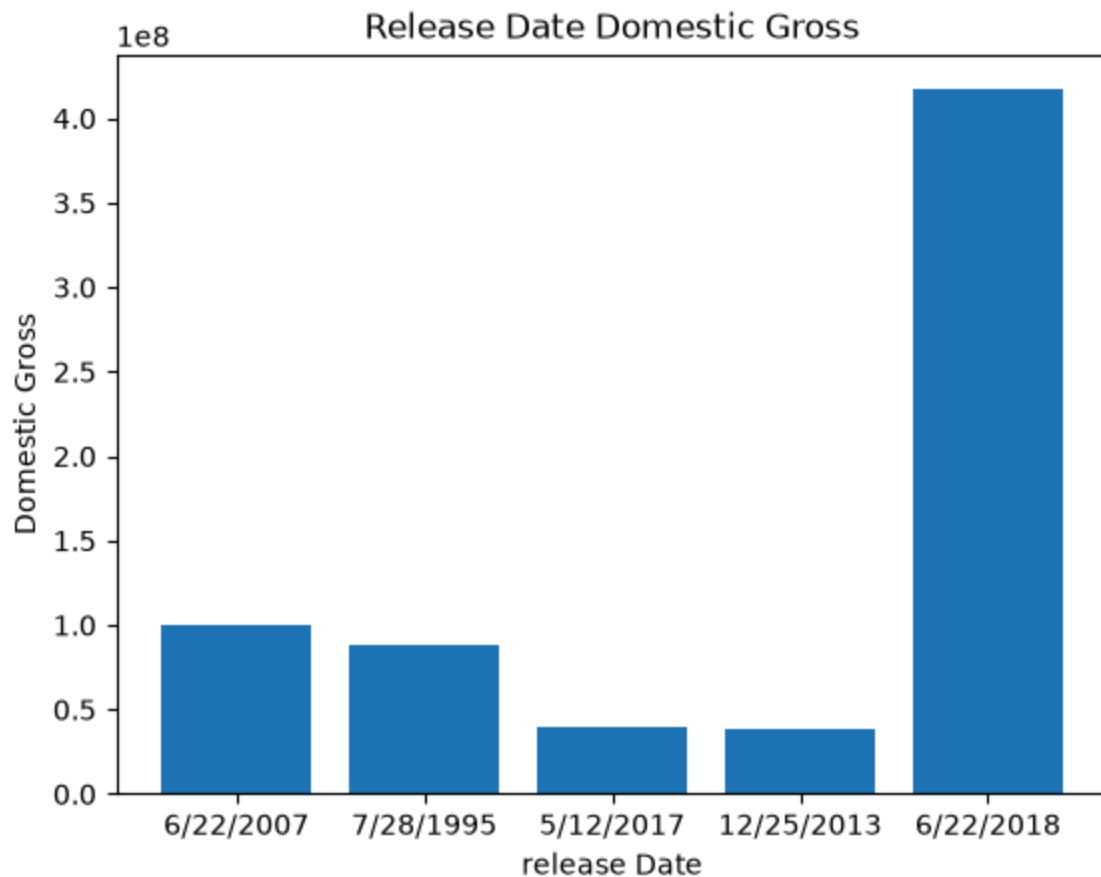
```
In [49]: df.groupby('mpaa_rating')['domestic_gross'].mean().sort_values(ascending
```

```
Out[49]: mpaa_rating
G      8.253156e+07
PG     6.742011e+07
PG-13  5.487091e+07
R      3.208529e+07
Name: domestic_gross, dtype: float64
```

```
In [45... plt.hist(df['worldwide_gross'])
plt.title('Distribution of Worldwide gross')
plt.xlabel('Worldwide Gross')
plt.ylabel('Frequency')
plt.show()
```



```
In [50... plt.bar(df['release_date'].head(), df['domestic_gross'].head())
plt.title('Release Date Domestic Gross')
plt.xlabel('release Date')
plt.ylabel('Domestic Gross')
plt.show()
```



```
In [54]: df.groupby('distributor')['worldwide_gross'].sum().sort_values(ascending=
```

```
Out[54]: distributor
Warner Bros.          4.821614e+10
20th Century Fox      4.697384e+10
Universal             4.599647e+10
Sony Pictures         3.747076e+10
Paramount Pictures    3.732193e+10
...
Morocco Junction Pi... 0.000000e+00
IFC Midnight          0.000000e+00
Hannover House        0.000000e+00
Lionsgate Premiere    0.000000e+00
Swipe Films           0.000000e+00
Name: worldwide_gross, Length: 186, dtype: float64
```

```
In [55]: df.groupby('genre')['production_budget'].mean()
```

```
Out[55]: genre
Action      5.850313e+07
Adventure   6.230153e+07
Comedy      2.517041e+07
Drama       2.238141e+07
Horror      1.881391e+07
Name: production_budget, dtype: float64
```

```
In [56... df.groupby('mpaa_rating')['domestic_gross'].median()
```

```
Out[56]: mpaa_rating
          G      58611764.5
          PG     48006762.0
          PG-13   34522221.0
          R      15934642.0
          Name: domestic_gross, dtype: float64
```

```
In [58... df.groupby(df['release_date'].str.slice(6,8))['movie'].count().sort_value
```

```
Out[58]: release_date
          00      820
          01      691
          20      537
          99      350
          19      218
          Name: movie, dtype: int64
```

```
In [59... df.groupby('genre')['domestic_gross'].max()
```

```
Out[59]: genre
          Action      416769345.0
          Adventure  474544677.0
          Comedy     285761243.0
          Drama      435110554.0
          Horror     327481748.0
          Name: domestic_gross, dtype: float64
```

```
In [60... df.groupby('mpaa_rating')['production_budget'].min()
```

```
Out[60]: mpaa_rating
          G      560000.0
          PG     300000.0
          PG-13   250000.0
          R      250000.0
          Name: production_budget, dtype: float64
```

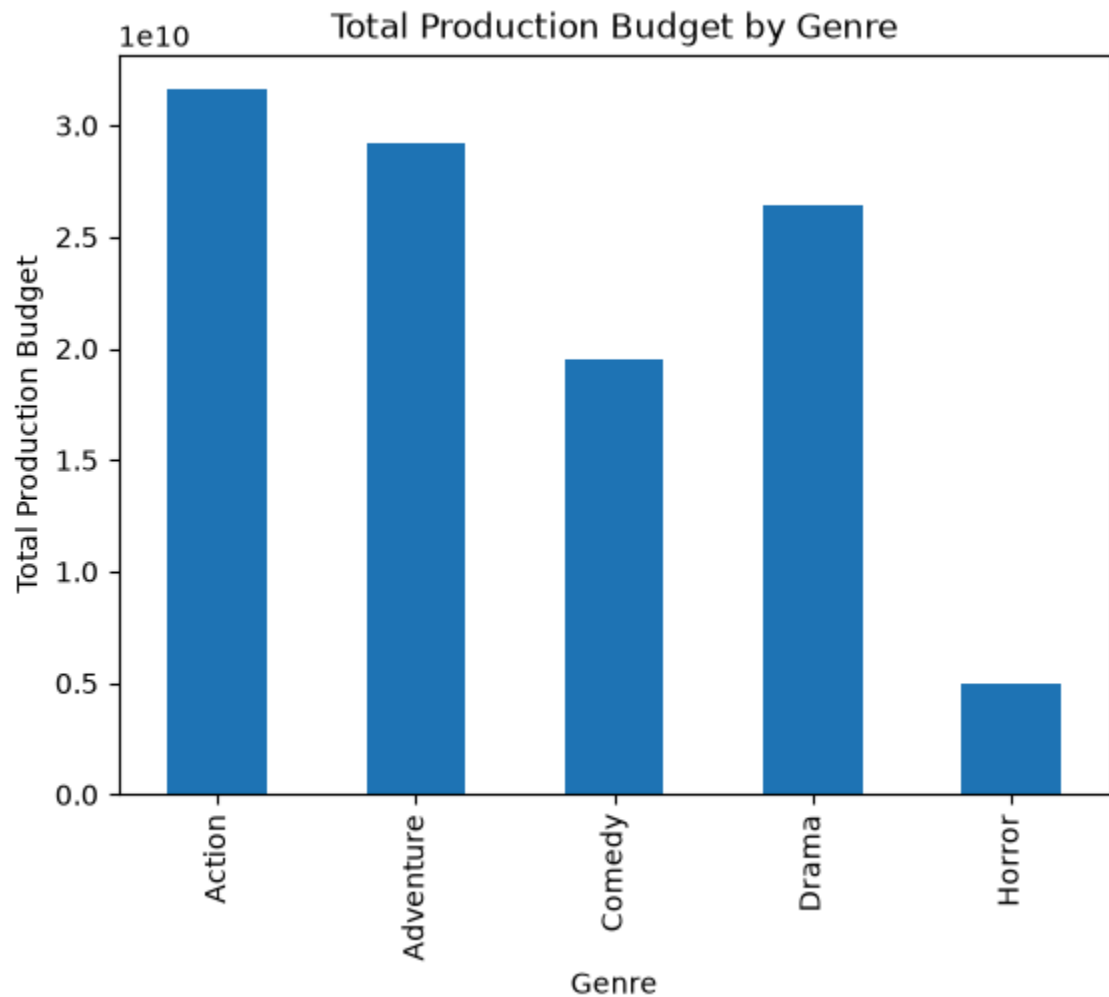
```
In [61... df.groupby('distributor')['worldwide_gross'].mean()
```

```
Out[61]: distributor
          20th Century Fox      1.726979e+08
          8X Entertainment      6.047691e+06
          A24                   2.332026e+07
          ARC Entertainment      2.248513e+06
          Abramorama Films      5.491850e+05
          ...
          Well Go USA           1.573010e+08
          Wrekin Hill Enterta...  6.155376e+07
          XLrator Media          9.270740e+05
          Yari Film Group Rel...  4.352631e+07
          Zeitgeist              6.173485e+06
          Name: worldwide_gross, Length: 186, dtype: float64
```

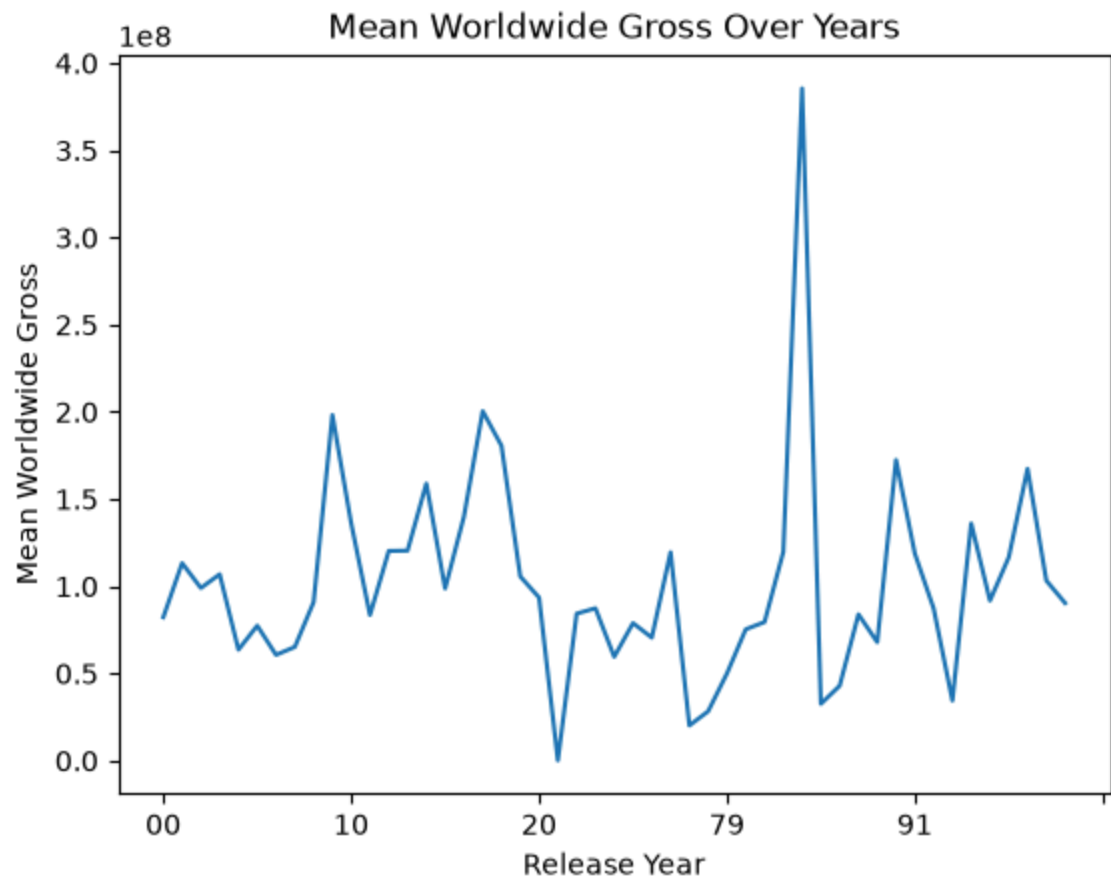
```
In [63... df.groupby('mpaa_rating')['worldwide_gross'].var()
```

```
Out[63]: mpaa_rating
G      3.875780e+16
PG     3.355641e+16
PG-13  2.422744e+16
R      9.384744e+15
Name: worldwide_gross, dtype: float64
```

```
In [64]: df.groupby('genre')['production_budget'].sum().plot(kind = 'bar')
plt.title('Total Production Budget by Genre')
plt.xlabel('Genre')
plt.ylabel('Total Production Budget')
plt.show()
```

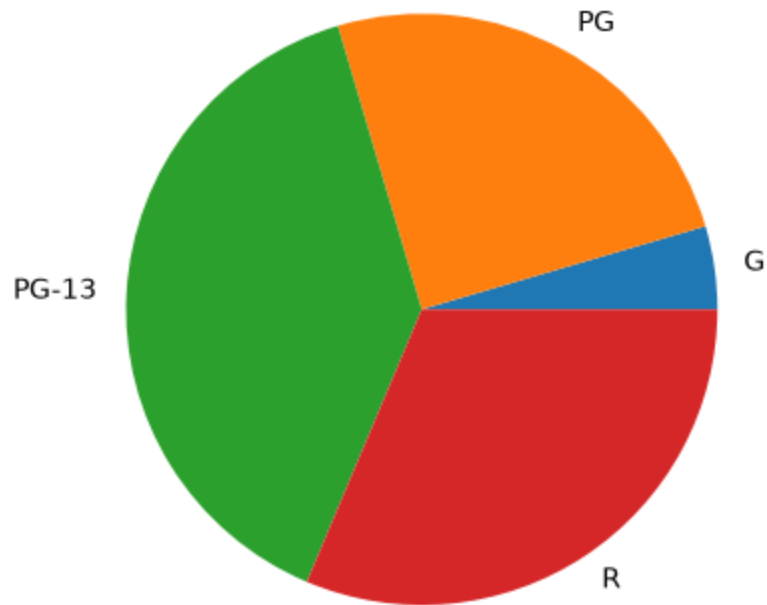


```
In [69]: df.groupby(df['release_date'].str.slice(6,8))['worldwide_gross'].mean().p
plt.title('Mean Worldwide Gross Over Years')
plt.xlabel('Release Year')
plt.ylabel('Mean Worldwide Gross')
plt.show()
```

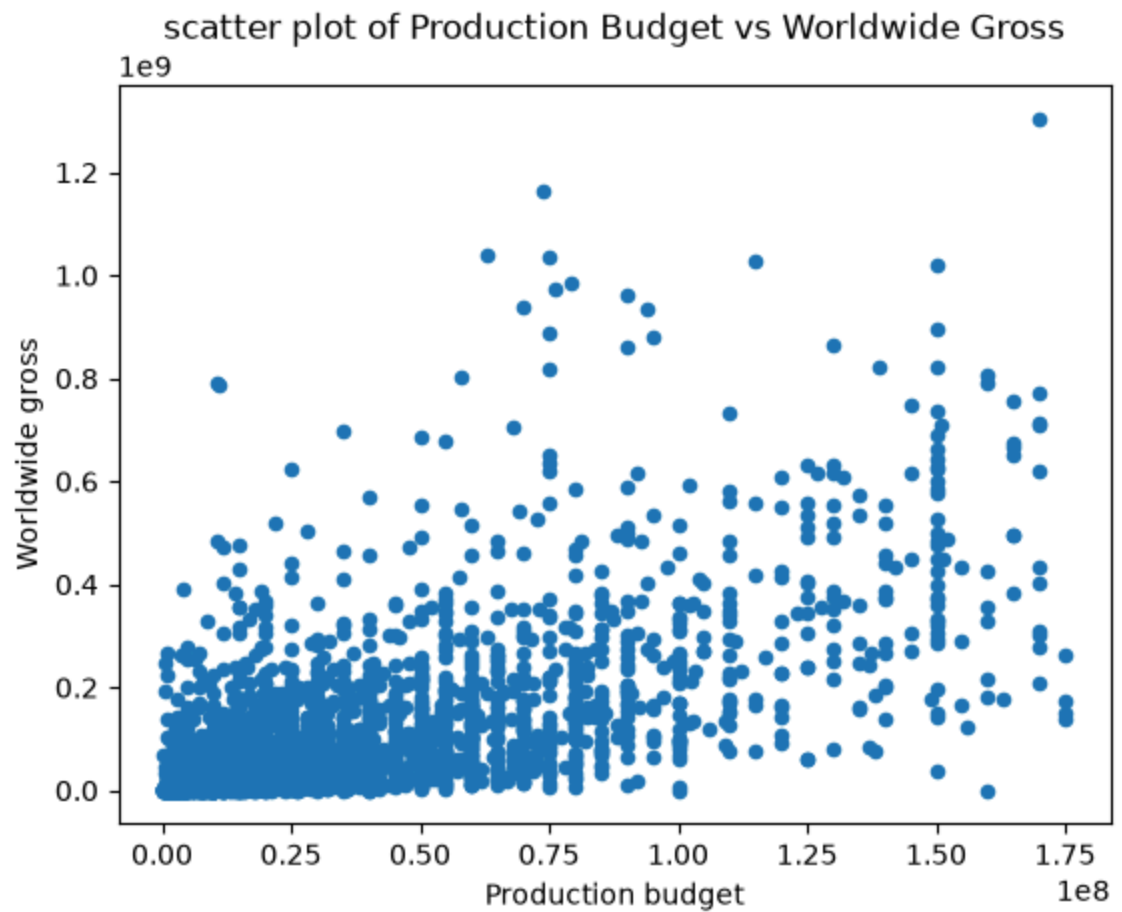


```
In [71... df.groupby('mpaa_rating')['domestic_gross'].sum().plot(kind='pie')
plt.title('Total Domestic Gross by Rating')
plt.ylabel("")
plt.show()
```

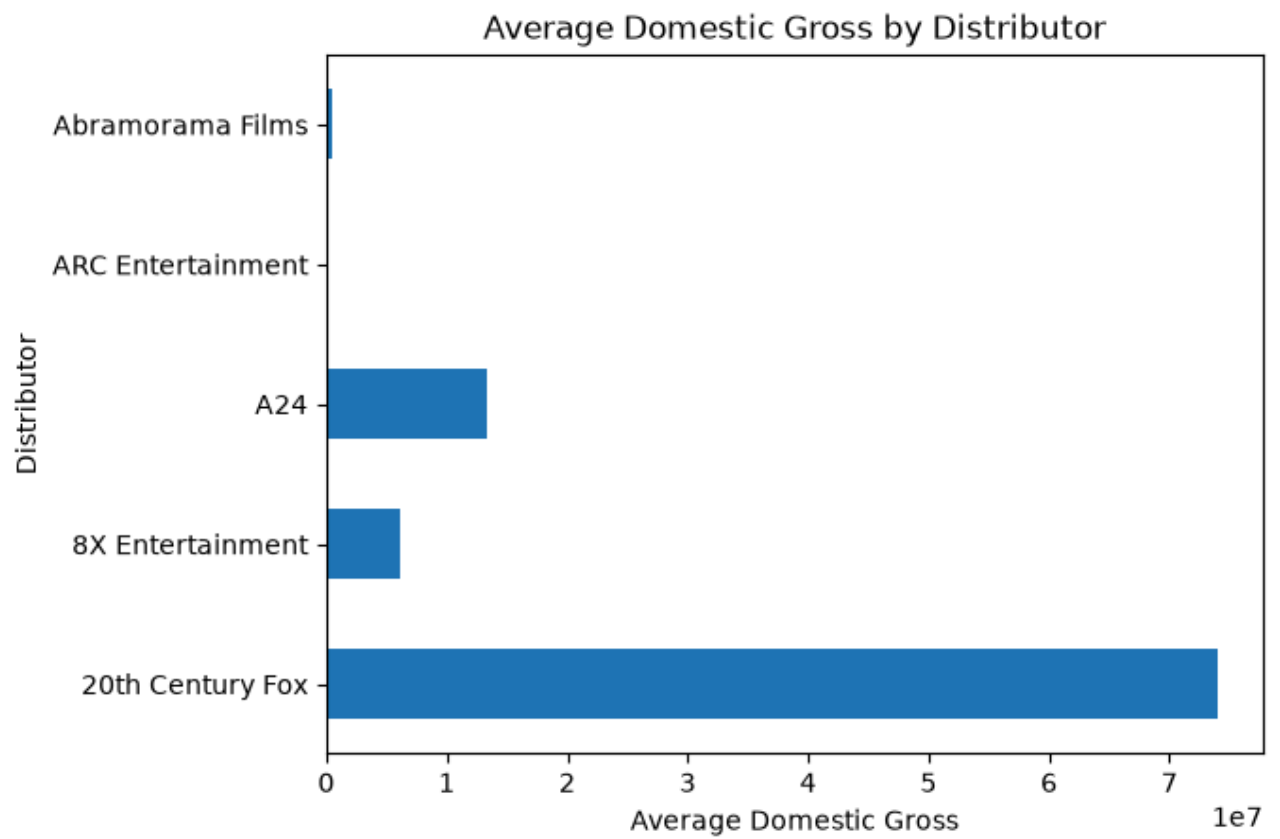
Total Domestic Gross by Rating



```
In [77... df.plot(kind='scatter', x='production_budget', y='worldwide_gross')
plt.title('scatter plot of Production Budget vs Worldwide Gross')
plt.xlabel('Production budget')
plt.ylabel('Worldwide gross')
plt.show()
```



```
In [76... df.groupby('distributor')['domestic_gross'].mean().head().plot(kind = 'bar')
plt.title('Average Domestic Gross by Distributor')
plt.xlabel('Average Domestic Gross')
plt.ylabel('Distributor')
plt.show()
```



In [ ]: